Current Controller Design in Discrete-time Domain for PMSM/G in Flywheel Energy Storage Systems

Xiang Zhang, Yves Mollet, Jiaqiang Yang, Johan Gyselinck

Abstract **–Wide speed range operation of the Permanent-Magnet Synchronous Motor/Generator (PMSM/G) in a Flywheel Energy Storage System (FESS) leads to lower sampling-to-fundamental frequency ratio than other PMSM applications. Oscillatory or unstable response can occur if the current controller design does not properly incorporate the discrete nature of PWM modulation method. A current controller directly designed in discrete-time domain is presented in this paper based on an accurate discrete model expressed with complex-vector, and some modifications and approximations are made to extend its application to salient machines. The digital delays are taken into consideration in the accurate discrete model and the coupling problem between d-q axis current loops is well solved. At last, the designed controller is validated by simulations and experiments on a PMSM/G test bench.**

***Index Terms*—Current controller design, Discrete-time domain, PMSM/G, FESS**

I. INTRODUCTION

Flywheel Energy Storage System (FESS) is a mechanical energy storage system where the energy is stored as the rotational kinetic energy of a flywheel, which is driven by an integrated motor/generator to interchange with electrical energy through a bidirectional power converter by repeated acceleration and deceleration [1]. Its application has been the research hotspots in the areas of uninterruptible power supplies (UPS), micro grids, wind power plants, rail transits, hybrid vehicles, etc [2-5] due to its higher energy transfer efficiency, higher instantaneous power, faster dynamic response, longer cycle lifetime and lower pollution to environment over other energy storage units [6,7].

Permanent Magnet Synchronous Motors/Generators (PMSM/Gs) are widely used in FESSs thanks to high efficiency, high power density, high reliability, and easy implementation of dual-directional power flow. However, PMSM/Gs in FESSs require wide speed range operation so that the sampling-to-fundamental-frequency ratio is often lower than other PMSM/G applications. At such low ratios, oscillatory or unstable response can occur if the current controller design does not properly incorporate the discrete nature of the PWM modulation method, mainly the digital delays between the sampling of the variables and the implementation of the reference signals [8].

Normally, the current controller is designed with a continuous frequency response method in the continuous-time domain, and transformed into its discrete form using an approximation method, such as the Euler forward difference and the Tustin transformation. The digital delays including sampling, computational and implementation delays can be taken into consideration by adding an additional first-order low-pass filter with a time constant of $3T_s/2$ in the controller design [9], but this approach is rather rough and is only feasible when the digital delays are small with respect to the fundamental period of the electrical machines [10]. However, when the digital delays are not negligible, a more accurate and complex model is needed, such as higher order Padé approximations in [11] and [12], and the phase and magnitude errors produced during the sampling period should be taken into consideration. Especially for the d-q axis current loop, a simple compensation of cross-coupling voltage based on feedback current cannot fully realize the decoupling. To this end, an accurate discrete-time domain model of PMSM needs to be built and the current controller should be directly designed in the discrete-time domain.

A current controller directly designed in discrete-time domain is presented in this paper based on an accurate discrete current loop model expressed with complex vector [8,13]. The complex vector allows to evaluate the performance and stability of a closed-loop system in continuous time and discrete time by means of classical analysis tools in time and frequency domains [14]. However, it cannot be directly applied to salient machines. Thus some modifications and approximations are made in this paper to extend its application to salient machines. The digital delays are taken into consideration in the accurate model and the coupling problem between d-q axis current loops is well solved. The fundamental principle and design process is elaborated in Section II. In Section III and Section IV, the performance of the current controllers designed in continuous-time domain and directly designed in discrete-time domain are compared through simulation and experimental results. At last, a conclusion is drawn in Section V.

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II. DESIGN OF THE CURRENT CONTROLLER IN DISCRETE-TIME DOMAIN

Fig. 1 presents the simplified schematic of the FESS studied in this paper, which consists of a flywheel coupled to a PMSM/G, a bidirectional converter, and a DC link.

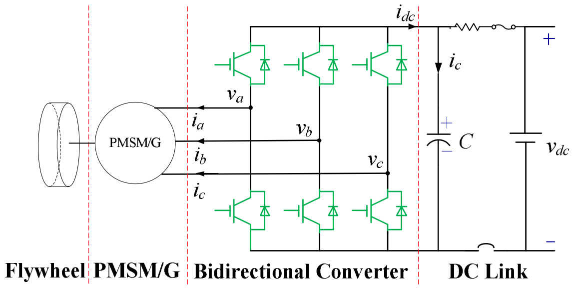


Fig.1 Schematic of the FESS

The model of a non-salient PMSM in the stationary frame and the rotor frame are expressed as

$$v_{\alpha\beta} = R_s i_{\alpha\beta} + L_s \frac{di_{\alpha\beta}}{dt} + e_{\alpha\beta} \quad (1)$$

and $$v_{dq} = R_s i_{dq} + L_s \frac{di_{dq}}{dt} + j\omega L_s i_{dq} + j\omega\psi_f \quad (2),$$

respectively, where $v_{\alpha\beta} = v_\alpha + jv_\beta$, $i_{\alpha\beta} = i_\alpha + ji_\beta$, $v_{dq} = v_d + jv_q$ and $i_{dq} = i_d + ji_q$ are the voltage and current vectors. ψ_f is the flux linkage of the permanent magnets. θ and ω are the electrical angular position and velocity, respectively. R_s and L_s are the phase resistance and inductance, respectively. $e_{\alpha\beta} = -\omega\psi_f \sin\theta + j\omega\psi_f \cos\theta$ is the back EMF vector in the stationary frame.

When developing the discrete-time domain model of the PMSM, a zero-order holder is used for each variable. As the input voltage is modulated by SVPWM strategy in the stationary frame, the amplitude and phase of v_α, v_β are constant during one sampling period. it is more appropriate to model the input voltage in the stationary frame with a zero-order holder.

However, the vector control strategy is normally developed in the rotor frame as the back EMF is constant in the rotor frame. Therefore, it is advisable to model the back EMF in the rotor frame. For this end, the superposition principle is applied to separately model the input voltage and the back EMF as follows.

A. Discrete-time domain current loop model of PMSM only excited by the input voltage

Only considering the excitation of the input voltage, the voltage equations of the PMSM stationary frame (2) in the continuous-time domain is simplified as

$$v_{\alpha\beta} = R_s i_{\alpha\beta} + L_s \frac{di_{\alpha\beta}}{dt} \quad (3).$$

Solving the differential equation (3) and considering the digital delay of one sampling period T_s, the discrete-time domain current loop model of PMSM is derived as

$$i_{\alpha\beta}[k] = i_{\alpha\beta}[k-1]e^{-R_sT_s/L_s} + v_{\alpha\beta}[k-2]\frac{1-e^{-R_sT_s/L_s}}{R_s} \quad (4).$$

Substituting the relationships $i_{\alpha\beta}[k]\cdot e^{-j\theta[k]} = i_{dq}[k]$ and $v_{\alpha\beta}[k]\cdot e^{-j\theta[k]} = v_{dq}[k]$ into (4), the discrete-time domain current loop dynamic of the PMSM only excited by the input voltage is derived in the rotor frame as

$$\begin{aligned} i_{dq}[k] &= i_{dq}[k-1]e^{-j\omega T_s}e^{-R_sT_s/L_s} \\ &+ v_{dq}[k-2]e^{-j2\omega T_s}\frac{1-e^{-R_sT_s/L_s}}{R_s} \end{aligned} \quad (5).$$

B. Discrete-time domain current loop model of PMSM excited only by the back EMF

When only considering the back EMF, the dynamics of the PMSM in rotor frame is expressed as

$$L_s \frac{di_{dq}}{dt} = -(R_s + j\omega L_s)i_{dq} - j\omega\psi_f \quad (6).$$

Solving the differential equation (6), the discrete-time domain solution is expressed as

$$\begin{aligned} i_{dq}[k] &= i_{dq}[k-1]e^{-(R_s+j\omega L_s)T_s/L_s} \\ &- j\omega\psi_f \frac{1-e^{-(R_s+j\omega L_s)T_s/L_s}}{R_s + j\omega L_s} \end{aligned} \quad (7).$$

C. Integrated discrete-time domain current loop model of PMSM excited by both input voltage and back EMF

Observing (5) and (7), it can be found out that the first terms on the right hand of the equations are the same, and the second terms are the dynamics excited by the input voltage and back EMF, respectively. Applying the superposition principle, the integrated discrete-time domain model of the PMSM considering both input voltage and back EMF is expressed as

$$\begin{aligned} i_{dq}[k] &= i_{dq}[k-1]e^{-(R_s+j\omega L_s)T_s/L_s} \\ &+ v_{dq}[k-2]e^{-j2\omega T_s}\frac{1-e^{-R_sT_s/L_s}}{R_s} \\ &- j\omega\psi_f \frac{1-e^{-(R_s+j\omega L_s)T_s/L_s}}{R_s + j\omega L_s} \end{aligned} \quad (8),$$

where the last term can be regarded as a disturbance and canceled by feedforward compensation.

D. Design of the current controller

a) Design of current controller in continuous-time domain

The most commonly used current controller (9) in continuous domain is illustrated in Fig.2. A simple PI controller is applied, and the state feedback decoupling method is added to eliminate the cross coupling between d- and q-axes. The back EMF is compensated afterwards, so it is not included in (9). The state feedback decoupling is realized by using the instantaneous feedback $j\omega\hat{L}_s i_{dq}$ to cancel the cross-coupling term, which will obtain a decoupled plant model that can be regulated appropriately by the PI controller. However, this method cannot accurately decouple the model when transformed to discrete-time domain, as the digital delays are not accurately considered during the discretizing process.

$$v_{dq}^* = (i_{dq}^* - i_{dq})(K_p + \frac{K_i}{s}) + \mathrm{j}\omega\hat{L}_s i_{dq} \quad (9)$$

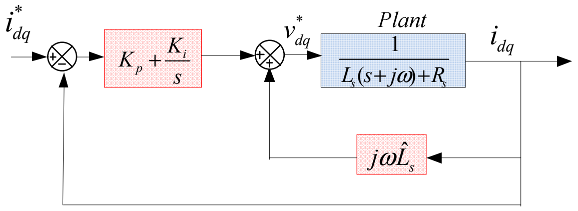


Fig.2 Block diagram of the PI current controller designed in continuous-time domain with state feedback decoupling

b) Design of current controller designed in discrete-time domain

Assuming the last term of (8) is perfectly compensated, the discrete-time domain transfer function from the input voltage to the output current is expressed as

$$\begin{aligned} G_p(z) &= \frac{i_{dq}(z)}{v_{dq}(z)} \\ &= \frac{1}{R_s}\frac{e^{-j2\omega Ts}z^{-2}(1-e^{-R_sT_s/L_s})}{1-e^{-(R_s+j\omega L_s)T_s/L_s}z^{-1}} \quad (10), \\ &\triangleq \frac{z^{-1}B(z)}{A(z)} \end{aligned}$$

where $A(z)$ and $B(z)$ are the denominator and numerator of the plant, respectively. $G_p(z)$ includes a complex pole $p = e^{-(R_s + j\omega L_s)T_s/L_s}$, which moves with the speed variation.

Applying zero-pole pair cancellation strategy, a complex zero $z = e^{-(R_s+j\omega L_s)T_s/L_s}$ is incorporated in the control law as

$$\begin{aligned} v_{dq}^* &= (i_{dq}^* - i_{dq})\frac{K\hat{A}(z)}{1-z^{-1}} \\ &= (i_{dq}^* - i_{dq})\frac{Ke^{j2\omega Ts}\left(1-e^{-(R_s+j\omega L_s)T_s/L_s}z^{-1}\right)}{1-z^{-1}} \end{aligned} \quad (11),$$

where $\hat{A}(z)$ is the estimated $A(z)$, and K is the gain of the controller.

The block diagram of the current controller designed in discrete-time domain is presented as Fig.3.

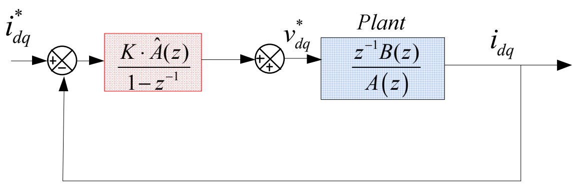


Fig.3 Block diagram of the current controller designed in discrete-time domain with zero-pole pair cancelation

E. Performance Analysis

Considering the PMSM current loop model in discrete-time domain (10), the current controller designed in continuous-time domain (9) is transformed to discrete-time domain as

$$v_{dq}^* = (i_{dq}^* - i_{dq})\left(K_p + \frac{K_i}{1-z^{-1}}\right) + \mathrm{j}\omega\hat{L}_s G_p(z) v_{dq}^* \quad (12),$$

which can be further transformed to

$$v_{dq}^* = (i_{dq}^* - i_{dq})\left(K_p + \frac{K_i}{1-z^{-1}}\right)\frac{1}{1-\mathrm{j}\omega\hat{L}_s G_p(z)} \quad (13).$$

Substituting the $G_p(z)$ into (13), the real discrete closed loop transfer function for the current controller designed in continuous-time domain is derived as

$$\begin{aligned} H_{cl}(z) &= \frac{\left(K_p + \frac{K_i}{1-z^{-1}}\right)G_p(z)}{1+\left(K_p + \frac{K_i}{1-z^{-1}} - \mathrm{j}\omega\hat{L}_s\right)G_p(z)} \\ &= \frac{\frac{1}{R_s}\left(K_p + \frac{K_i}{1-z^{-1}}\right)e^{-j2\omega T_s}z^{-2}\left(1-e^{-R_sT_s/L_s}\right)}{\left(1-e^{-(R_s+j\omega L_s)T_s/L_s}z^{-1}\right)+\frac{1}{R_s}\left(K_p + \frac{K_i}{1-z^{-1}} - \mathrm{j}\omega\hat{L}_s\right)e^{-j2\omega T_s}z^{-2}\left(1-e^{-R_sT_s/L_s}\right)} \end{aligned} \quad (14)$$

Applying Taylor's Formula and keep the first two terms, (14) is transformed to

$$H_{cl}(z) = \frac{\left(K_p + \frac{K_i}{1-z^{-1}}\right)\frac{T_s}{L_s}e^{-j2\omega T_s}z^{-2}}{1-\left(1-\frac{R_sT_s}{L_s} - j\omega T_s\right)z^{-1} + \left(K_p + \frac{K_i}{1-z^{-1}} - \mathrm{j}\omega\hat{L}_s\right)\frac{T_s}{L_s}e^{-j2\omega T_s}z^{-2}} \quad (15),$$

where the complex cross coupling term is not completely cancelled due to the delays.

As for the current controller (11) designed in discrete-time domain, the closed transfer function is derived as

$$H_{cl}(z) = \frac{K(1-e^{-R_sT_s/L_s})z^{-2}}{R_s(1-z^{-1}) + K(1-e^{-R_sT_s/L_s})z^{-2}} \quad (16),$$

where the complex pole moving with the speed variation is completely cancelled so that the cross-coupling problem is well solved.

F. Implementation of the current controller in dSPACE

Since the parameters of the MATLAB/SIMULINK transfer function block cannot be complex, the control law is expressed as four separate transfer functions with real parameters.

For non-salient PMSMs, the transfer functions are expressed as

$$\begin{aligned} \frac{v_d^*(z)}{i_d^*(z)-i_d(z)} &= K\frac{\cos(2\omega T_s)z - e^{-R_sT_s/L_s}\cos(\omega T_s)}{z-1} \\ \frac{v_q^*(z)}{i_d^*(z)-i_d(z)} &= -K\frac{\sin(2\omega T_s)z - e^{-R_sT_s/L_s}\sin(\omega T_s)}{z-1} \\ \frac{v_d^*(z)}{i_q^*(z)-i_q(z)} &= K\frac{\sin(2\omega T_s)z - e^{-R_sT_s/L_s}\sin(\omega T_s)}{z-1} \\ \frac{v_q^*(z)}{i_q^*(z)-i_q(z)} &= K\frac{\cos(2\omega T_s)z - e^{-R_sT_s/L_s}\cos(\omega T_s)}{z-1} \end{aligned} \quad (17).$$

The parameter K can be tuned according to the required bandwidth of the current loop by $K=2\pi f_{cl}L_s$, where f_{cl} is the closed-loop bandwidth of the current loop.

As for salient PMSMs (L_q>L_d), the model cannot be directly expressed with the complex vector form. The most accurate way is to model it with a matrix formulation, which results in great increase of model complexity and computational burden. A method based on the flux dynamics is proposed in [10], which draws the conclusion that different parameters $K_d=2\pi f_{cl}L_d$ and $K_q=2\pi f_{cl}L_q$ should be separately set to ensure the same dynamic for d-q-axis current controllers. However, the essence of this method is to neglect the factor $e^{-R_s T_s / L_s}$ in (17) to derive

$$\begin{aligned}
\frac{v_d^*(z)}{i_d^*(z)-i_d(z)} &= K_d \frac{\cos(2\omega T_s)z-\cos(\omega T_s)}{z-1} \\
\frac{v_q^*(z)}{i_d^*(z)-i_d(z)} &= -K_q \frac{\sin(2\omega T_s)z-\sin(\omega T_s)}{z-1} \\
\frac{v_d^*(z)}{i_q^*(z)-i_q(z)} &= K_d \frac{\sin(2\omega T_s)z-\sin(\omega T_s)}{z-1} \\
\frac{v_q^*(z)}{i_q^*(z)-i_q(z)} &= K_q \frac{\cos(2\omega T_s)z-\cos(\omega T_s)}{z-1}
\end{aligned} \quad (18),$$

so that it is only applicable to high speed operation and oscillation may occur when it is applied at medium or low speed. To solve this problem, a compromise is made by applying

$$\begin{aligned}
\frac{v_d^*(z)}{i_d^*(z)-i_d(z)} &= K_d \frac{\cos(2\omega T_s)z-e^{-R_s T_s/L_s}\cos(\omega T_s)}{z-1} \\
\frac{v_q^*(z)}{i_d^*(z)-i_d(z)} &= -K_q \frac{\sin(2\omega T_s)z-e^{-R_s T_s/L_s}\sin(\omega T_s)}{z-1} \\
\frac{v_d^*(z)}{i_q^*(z)-i_q(z)} &= K_d \frac{\sin(2\omega T_s)z-e^{-R_s T_s/L_s}\sin(\omega T_s)}{z-1} \\
\frac{v_q^*(z)}{i_q^*(z)-i_q(z)} &= K_q \frac{\cos(2\omega T_s)z-e^{-R_s T_s/L_s}\cos(\omega T_s)}{z-1}
\end{aligned} \quad (19),$$

where $L_s=\frac{1}{2}(L_d+L_q)$. As T_s is very small and $e^{-R_s T_s/L_s}$ is close to 1, the error tolerance of using the average value of L_d and L_q is acceptable if the saliency is not too high.

III. Simulation Results

A comprehensive simulation is conducted to test the performance of the current controllers by means of two speed steps to generate a transient process of current loop. A PI controller is used for the speed loop. A more intuitive current-step test would have led to a speed transient and a possible entering of the field-weakening region due to the limited inertia and speed range of the PMSM. The parameters of the investigated machine, converter and controller used in the simulation and the experiment in Section IV are listed in TABLE I.

TABLE I
Parameters of the test bench

Rated Current (A)	10.6
Pole Pairs	3
Rated Speed (rpm)	1000
PM Flux Linkage (V.s)	0.512
Stator Resistance (Ω)	3.0
d-axis Inductance (H)	0.077
q-axis Inductance (H)	0.153
Total Inertia (kg·m²)	0.0071
DC-link Voltage (V)	360
Switching Frequency (kHz)	5
Sampling Frequency (kHz)	10
Closed Loop Bandwidth of the Current Loop (kHz)	1

The waveforms of the reference and measured speeds are shown in Fig.4 and Fig.5. The speed reference increases to 1000rpm from zero speed at 0.5s, then drops back to zero at 1.5s. The corresponding d-q-axis current waveforms are presented in Fig.6 to Fig.9. Fig.6 and Fig.7 present the q-axis current waveforms of the controller designed in continuous and discrete-time domain, respectively.

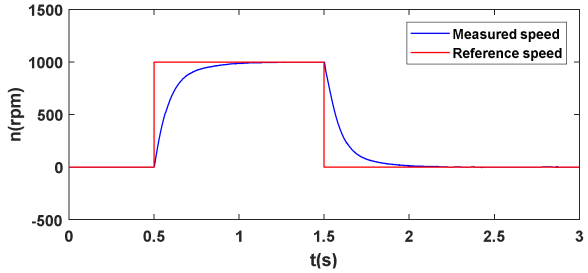


Fig.4 Speed step response of the current controller designed in continuous-time domain

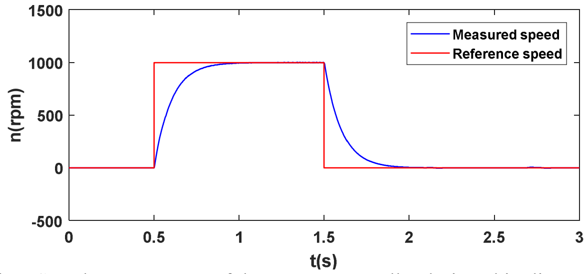
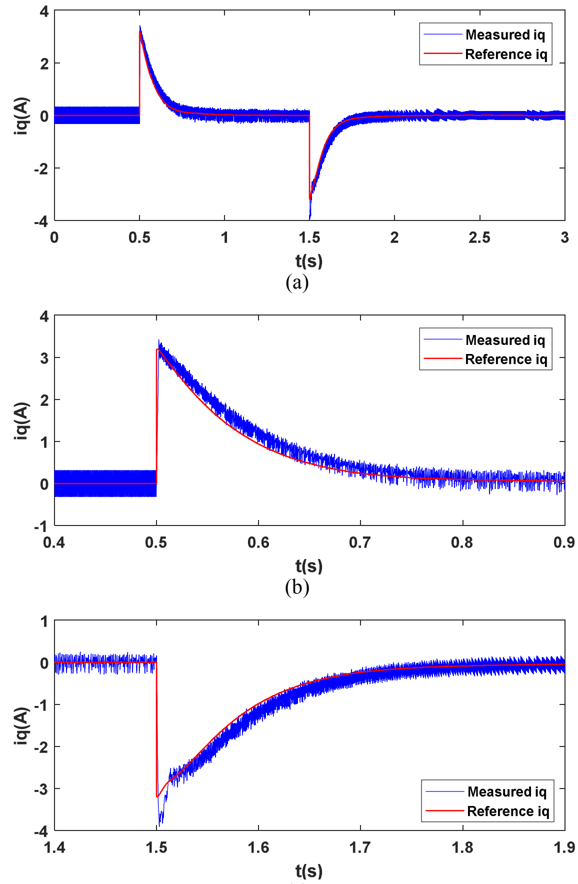


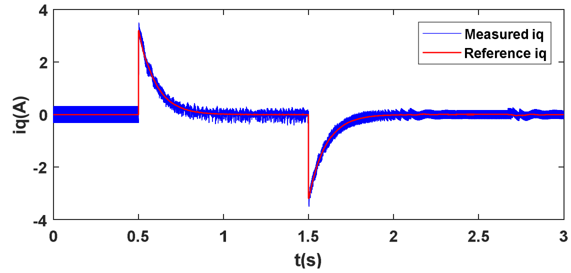
Fig.5 Speed step response of the current controller designed in discrete-time domain



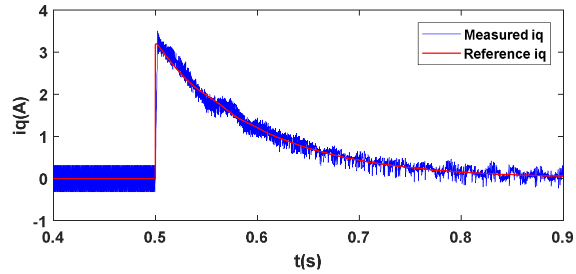
(a)
(b)
(c)
Fig.6 q-axis current of the current controller designed in continuous-time domain at the speed step

In general, the measured q-axis currents can track their references in steady state. However, a small tracking error exists in the transient process from the zoomed figures Fig.6 (b) to (c) for the controller designed in continuous-time domain. As for the controller directly designed in discrete-time domain, the q-axis current always track its reference with no error except for a tiny oscillation at the beginning.

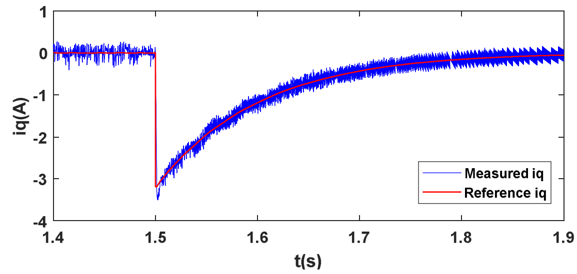
tracking and decoupling capability than the controller designed in the continuous time domain.



(a)



(b)



(c)

Fig.7 q-axis current of the current controller designed in discrete-time domain at the speed step

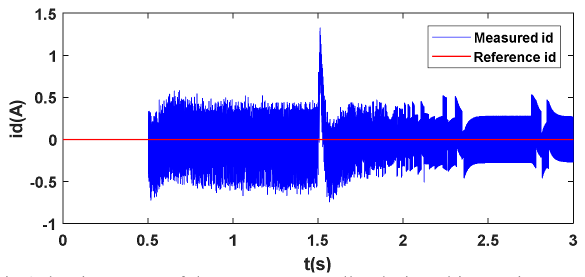


Fig.8 d-axis current of the current controller designed in continuous-time domain at the speed step

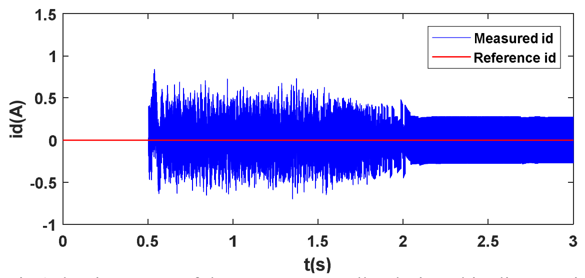


Fig.9 d-axis current of the current controller designed in discrete-time domain at the speed step

In Fig.8, the d-axis current of the controller designed in continuous time domain suffers from an obvious transient error at 1.5s when the speed reference jumps from 1000 rpm to 0 rpm. But for the controller designed in discrete-time domain, no obvious transient error occurs in Fig.9.

Therefore, the simulation results prove that the controller directly designed in the discrete-time domain has better tracking and decoupling capability than the controller designed in the continuous time domain.

IV. Experimental Results

The current controllers designed in the continuous- and discrete-time domains are experimentally tested on the test bench in Fig.10, which is based on dSPACE 1006 fast prototyping system and two coupled PMSMs. The performance of the current controllers is tested by giving several speed steps as shown in Fig.11 and Fig.12 to generate transient process of current loop. The speed reference firstly increases from the 100 rpm to 1000rpm, then drops to 500 rpm, again increases to 1000 rpm and drops back to 100 rpm in the end. The speed responses are not exactly the same due to uncertain disturbances. The corresponding current waveforms are presented in Fig.13 to Fig.16.

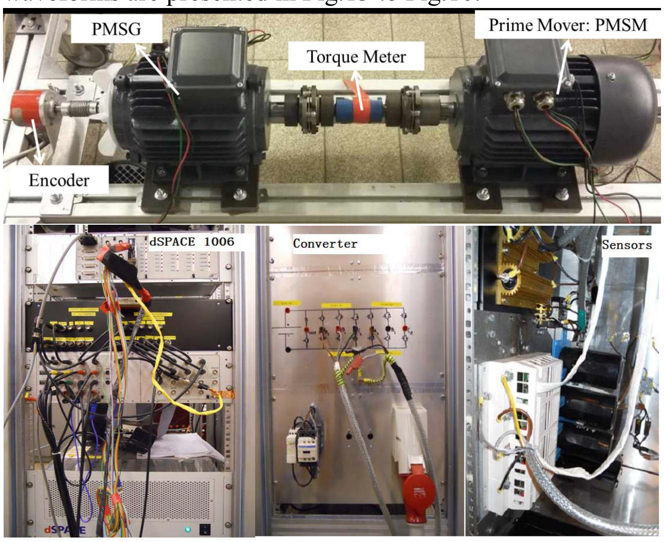


Fig.10 The PMSM test bench used in the experiment

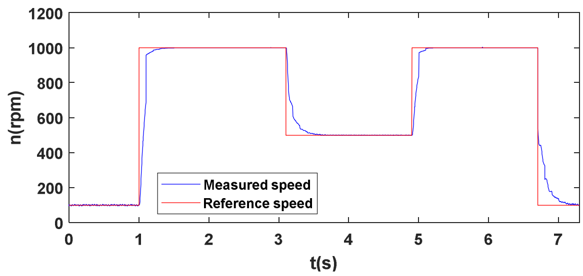


Fig.11 Speed step response of the current controller designed in continuous-time domain

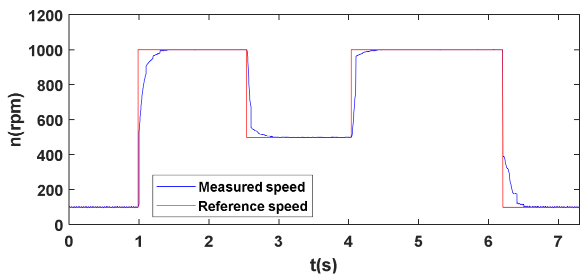
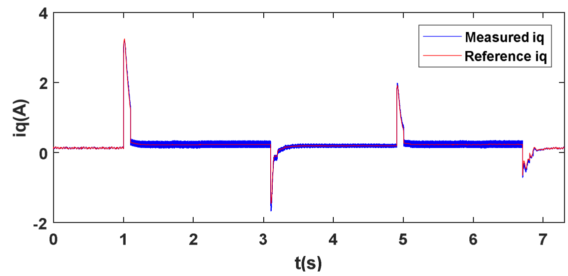


Fig.12 Speed step response of the current controller designed in discrete-time domain

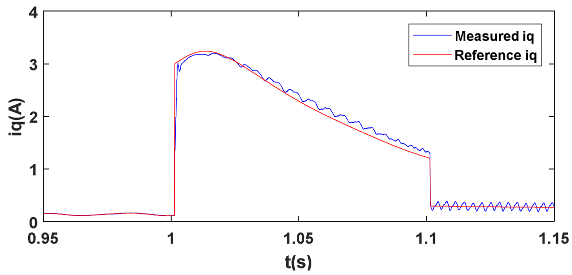
The waveforms of the q-axis current for the two controllers are presented as Fig.13 and Fig. 14. In both cases, the q-axis current well tracks the reference. However, it can be found in the zoomed figures Fig.14-(b) and Fig.14-(c) that the q-axis current of the controller designed in discrete-time domain well tracks the reference in transient process, whereas that of the controller designed in continuous-time domain in Fig.13-(b) and Fig.13-(c) suffers from a transient tracking error. It proves that the current controller designed in discrete-time domain has improved current tracking capability.

The d-axis current waveforms of the two current controllers designed in the continuous- and discrete-time domains are presented as Fig.15 and Fig.16, respectively. In Fig.15, an obvious transient error occurs in the d-axis current waveform of the current controller designed in continuous-time domain when the reference of the q-axis suddenly changes, even if the cross-coupling compensation is applied. But in Fig.16 the d-axis current performance is much better, no obvious transient error happens when the q-axis current reference suddenly changes. The comparison of Fig.15 and Fig.16 reveals that the current controller directly designed in the discrete-time domain has better decoupling performance than the current controller designed in continuous-time domain.

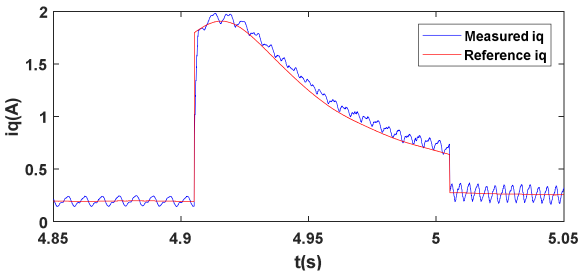
In general, the same conclusion can be drawn as the simulation results.



(a)

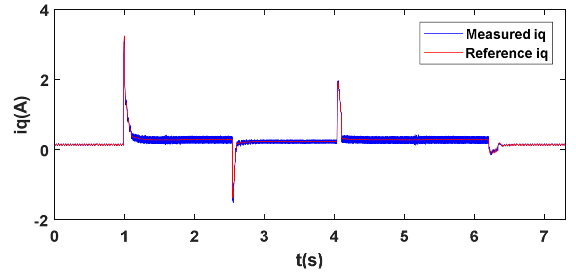


(b)

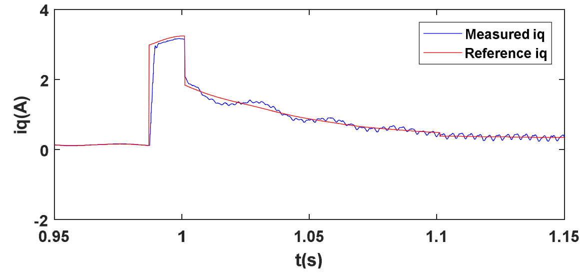


(c)

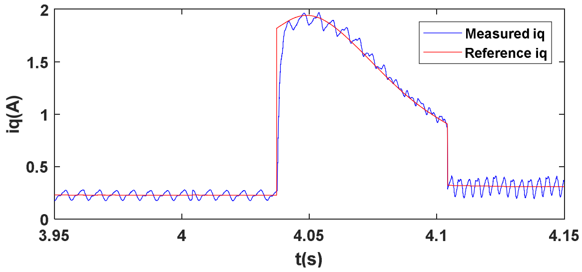
Fig.13 q-axis current of the current controller designed in continuous-time domain at the speed step



(a)



(b)



(c)

Fig.14 q-axis current of the current controller designed in discrete-time domain at the speed step

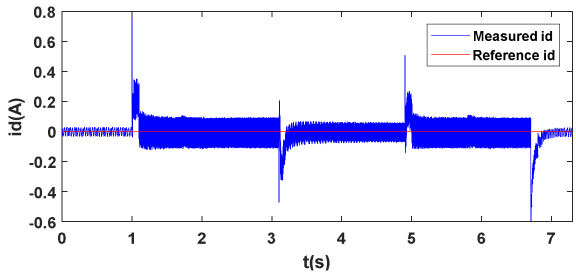


Fig.15 d-axis current of the current controller designed in continuous-time domain at the speed step

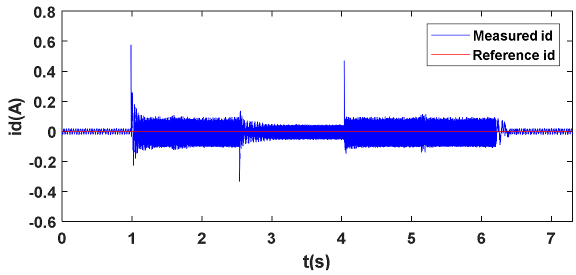


Fig.16 d-axis current of the current controller designed in continuous-time domain at the speed step

V. CONCLUSION

Wide speed range operation of the Permanent-Magnet Synchronous Motor/Generator (PMSM/G) in a Flywheel Energy Storage System (FESS) leads to lower sampling-to-fundamental frequency ratio than other PMSM applications. Oscillatory or unstable response can occur if the current controller design does not properly incorporate the discrete nature of PWM modulation method. A current controller directly designed in discrete-time domain is presented in this paper based on an accurate discrete current loop model expressed with complex-vector, and some modifications and approximations are made to extend its application to salient machines. The digital delays, including sampling, computational and implementation delays, are taken into consideration. The performance of the designed current controller is tested and compared with that of a current controller designed in continuous-time domain. The results prove that the current controller designed in discrete-time domain has better tracking and decoupling performance.

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Biographies

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